

RESOBOT

Enterprise Incident Management Platform — Feature Specification

ResoBot is a comprehensive, AI-driven incident management and resolution orchestration platform. Built on a modern MERN stack, it combines real-time data synchronization, headless robotic process automation (RPA), and advanced natural language processing to streamline corporate support operations.

1. INTELLIGENT AUTOMATION (AI & RPA)

OpenAI GPT-4o Document Parsing

Automatically processes unstructured incident files (PDFs, images, raw text) using advanced OCR and NLP. The AI extracts core issues, suggests priority levels, and determines optimal department routing before human review.

UiPath Orchestrator Integration

A direct API bridge to UiPath cloud environments. Admins can force-start headless bots to scan corporate drives (e.g., Google Drive) for new incident reports and automatically push them to the platform via secure webhooks.

Automated Notification Engine

Whenever a new incident is ingested via bot or manual upload, the system automatically dispatches priority-flagged email alerts to the assigned Department Supervisors via Nodemailer.

2. CORE INCIDENT OPERATIONS

The Incident Vault

A centralized, high-performance data grid storing all historical and active incidents. Features comprehensive filtering by status, department, and priority, along with full resolution tracking and audit trail capabilities.

AI Draft Builder (Manual Ingestion)

A drag-and-drop workspace allowing agents to upload corrupted or physical files. The system provides a side-by-side view of the raw extracted text and the AI's structured interpretation, allowing agents to correct data before committing to the Vault.

Real-Time Telemetry Dashboard

Powered by WebSockets (Socket.io), the dashboard provides live metrics on system SLA compliance, active incident volumes, and resolution times without requiring page refreshes.

3. ADMINISTRATION & SECURITY

Enterprise-Grade Authentication

Secure JSON Web Token (JWT) architecture with 30-day rolling expiration. Passwords are cryptographically hashed at the database layer using Bcrypt via Mongoose pre-save hooks.

Role-Based Access Control (RBAC)

Granular permissions system governing user access. Roles include Admin, Manager, Agent, and Viewer, dictating exactly what data can be mutated or viewed.

Team Directory & Provisioning

A dedicated administrative portal for managing workforce access. Admins can provision new accounts, set temporary passwords, assign departments, and instantly revoke system access by toggling account status.

Bot Telemetry & Error Failsafes

A dedicated Bot Control screen displaying execution history, terminal output, and Try/Catch visual error logs (including UI element screenshots) directly from the UiPath Orchestrator.

TECHNICAL SPECIFICATIONS

Frontend Architecture	React.js (Vite), React Router, Tailwind CSS, Lucide Icons, Context API
Backend Architecture	Node.js, Express.js, Socket.io (WebSockets)
Database & ORM	MongoDB Atlas, Mongoose ODM
API Integrations	OpenAI (GPT-4o), UiPath Orchestrator OData API
Security Stack	Helmet.js, Express Rate Limiter, Bcryptjs, JWT, CORS